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(54) **CABLE-STAYED JACKING REVERSING
TYPE SIXTEEN-WHEEL FOUR-WAY
SHUTTLE**

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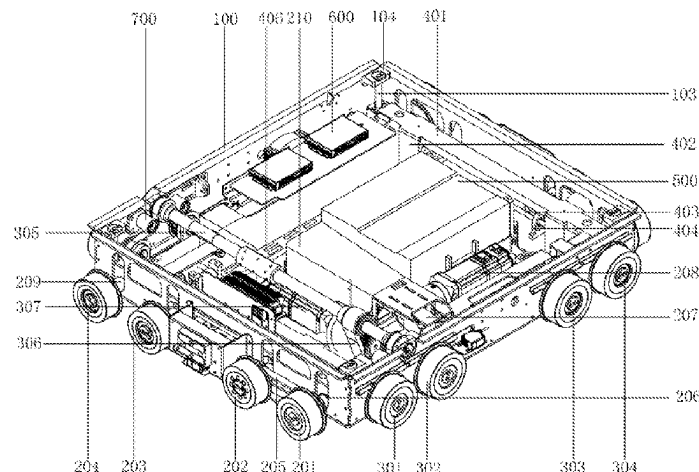
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(57) **ABSTRACT**

A cable-stayed jacking reversing type sixteen-wheel four-way shuttle comprises a supporting frame, a transverse movement drive mechanism, a longitudinal movement drive mechanism and a cable-stayed reversing jacking mechanism, wherein the transverse movement drive mechanism includes transverse driving wheels and first/second/third transverse driven wheels; the longitudinal movement drive mechanism includes first/second longitudinal driving wheels and first/second longitudinal driven wheels; and, the cable-stayed reversing jacking mechanism includes jacking transverse plates and cable-stayed jacking blocks, in which a ball bearing sleeve is movably arranged in a jacking shaft hole, a cable-stayed jacking shaft is arranged between the front and rear sides of the ball bearing sleeve, and the cable-stayed jacking blocks are connected with each other through ball screws. The shuttle is a sixteen-wheel two-way moving vehicle which is compact, cost-effective and space saving,

(Continued)



and which also greatly improves the carrying capacity and obstacle crossing ability.

10 Claims, 6 Drawing Sheets

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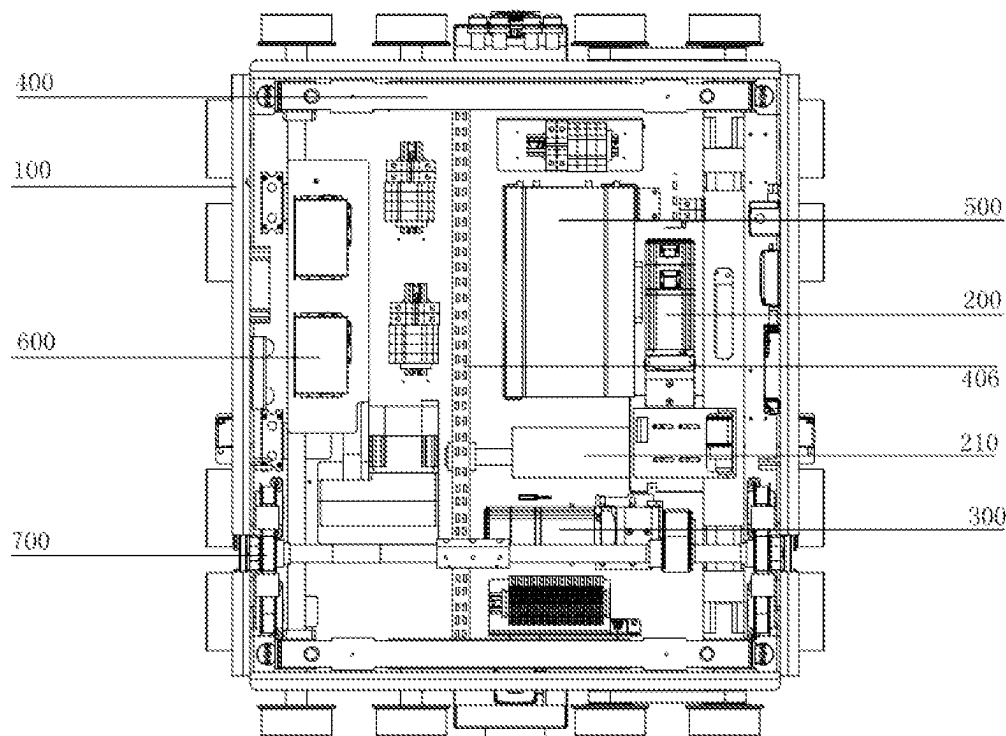


FIG. 1

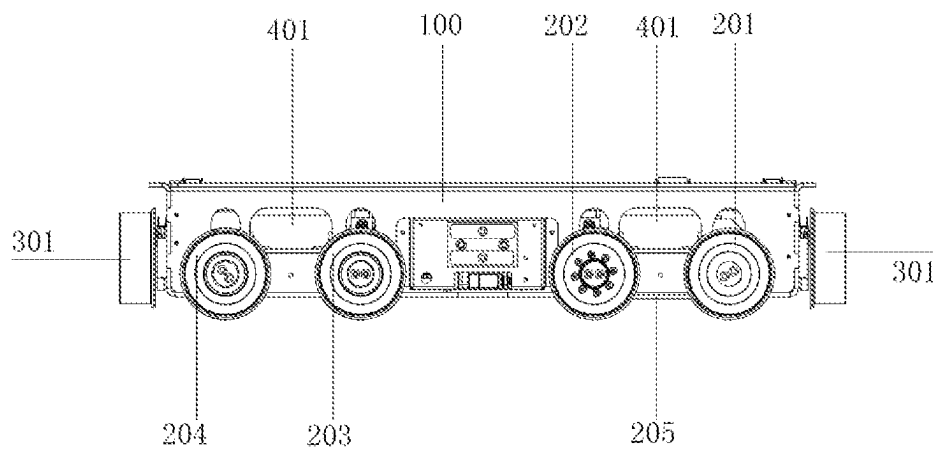


FIG. 2

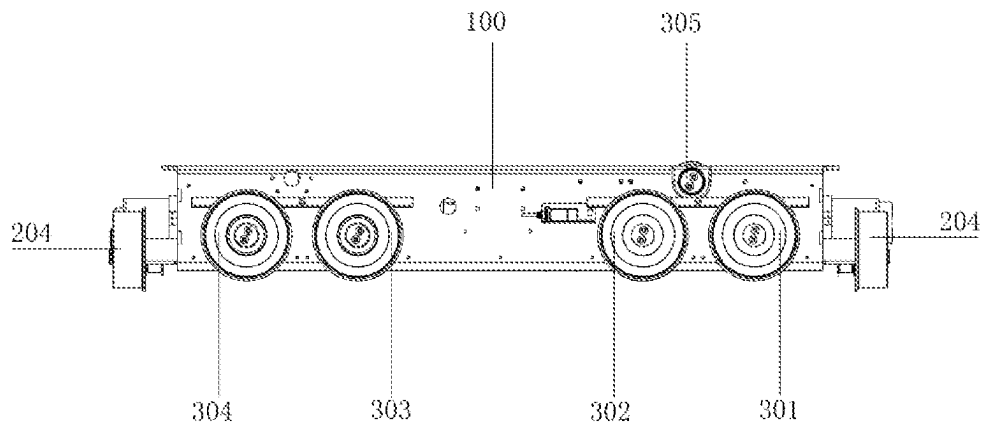


FIG. 3

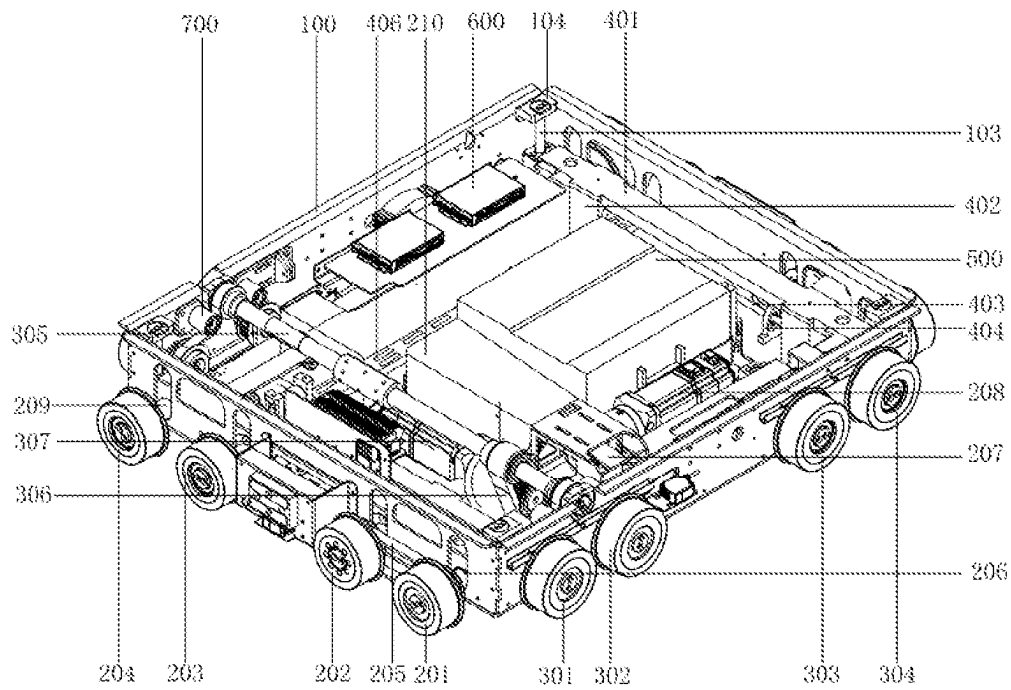


FIG. 4

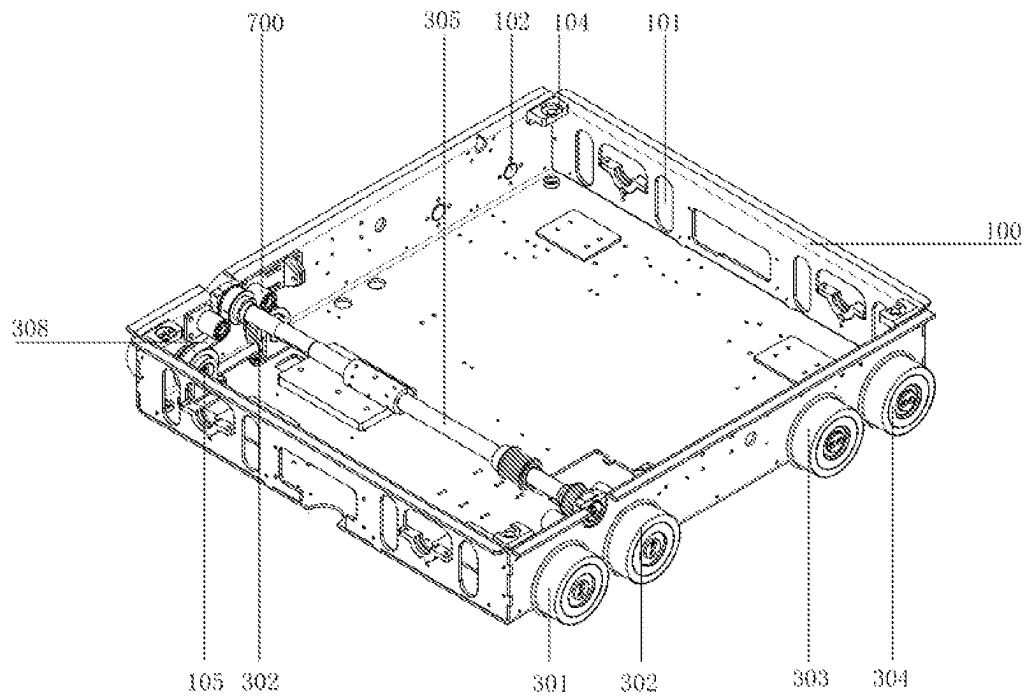


FIG. 5

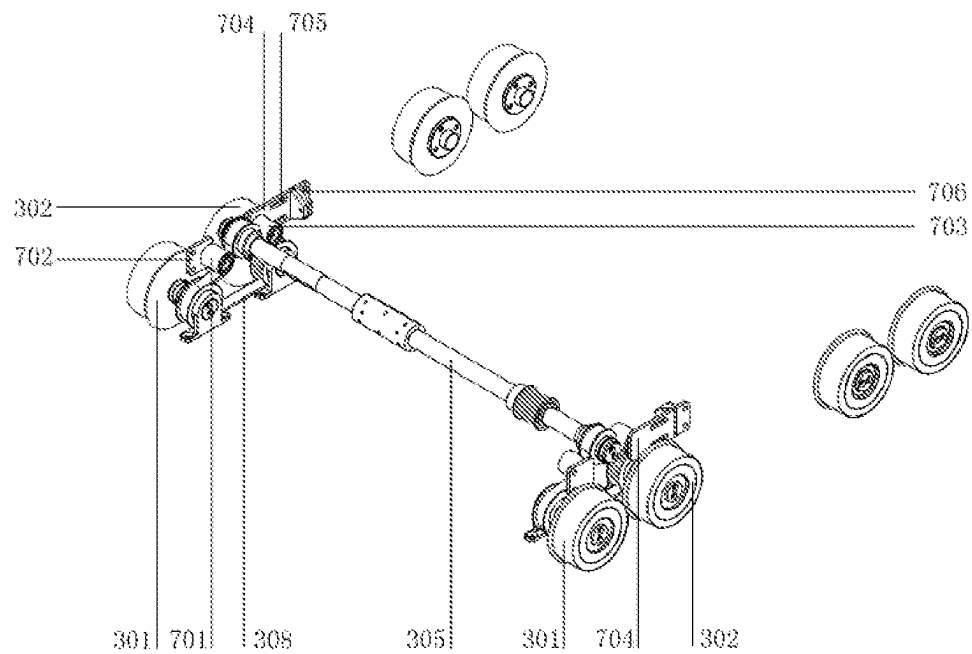


FIG. 6

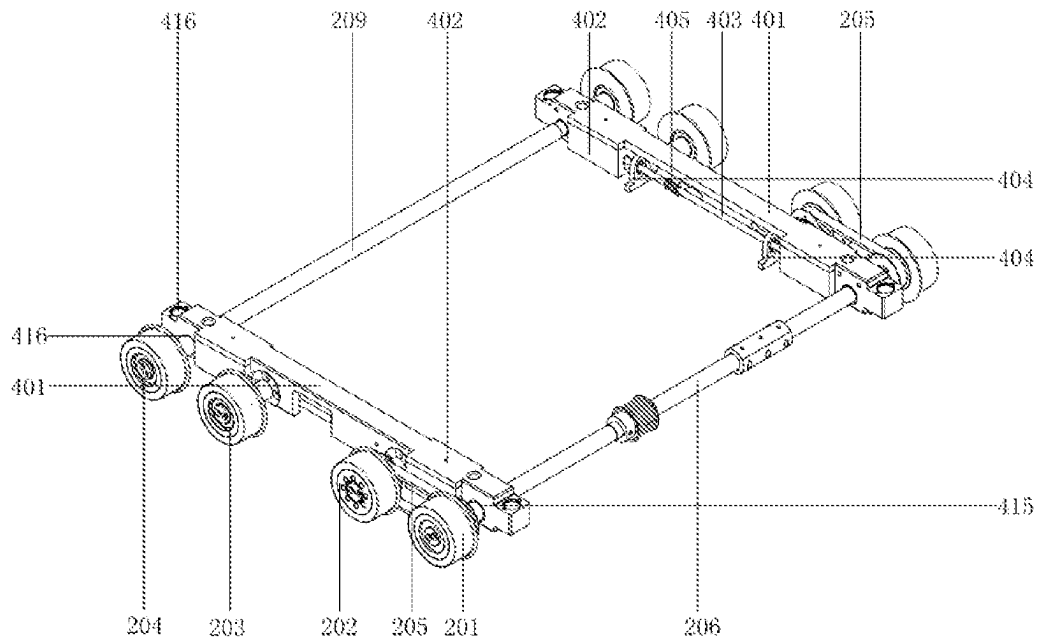


FIG. 7

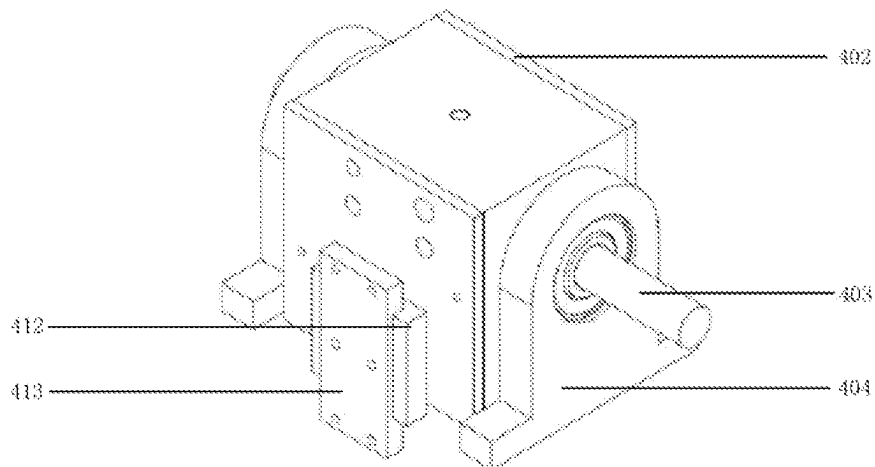


FIG. 8

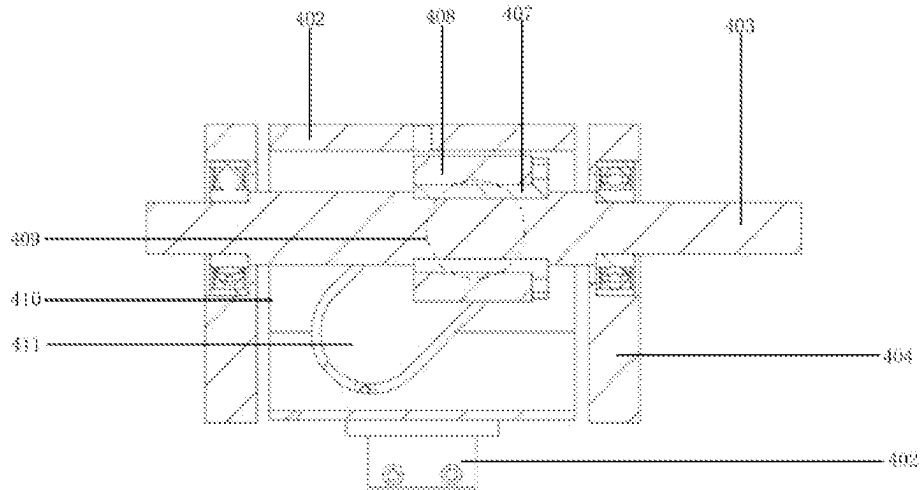


FIG. 9

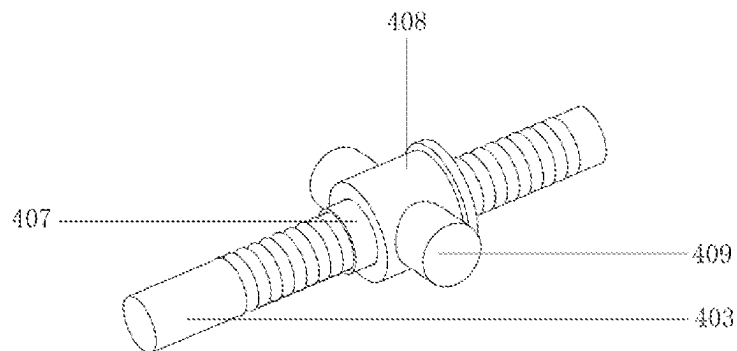


FIG. 10

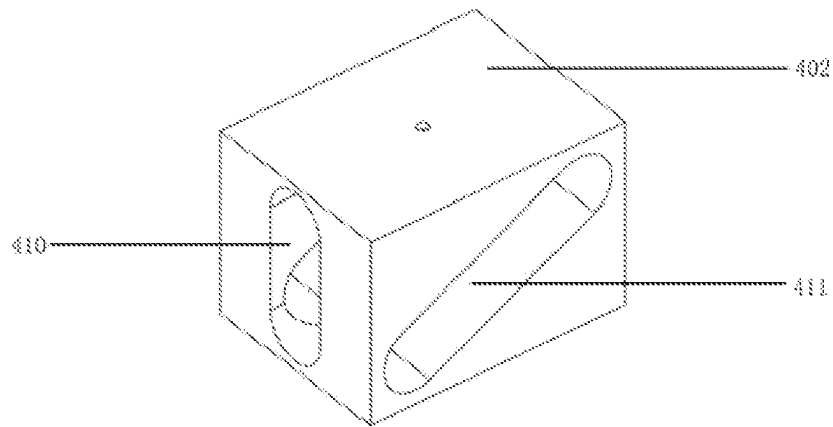


FIG. 11

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CABLE-STAYED JACKING REVERSING TYPE SIXTEEN-WHEEL FOUR-WAY SHUTTLE

TECHNICAL FIELD

The present disclosure relates to the technical field of shuttles, and in particular, to a cable-stayed jacking reversing type sixteen-wheel four-way shuttle.

BACKGROUND

With a rapid development of warehousing and logistics industry, a three-dimensional warehouse space with higher a utilization rate is required. For the three-dimensional warehouse, it is hoped that more shelves should be arranged in the horizontal direction and more shelf compartments should be arranged in the height direction, so as to minimize the space occupied by incoming and unloading passages.

The current three-dimensional warehouse structure prefers to use the type of one incoming and unloading lane and multiple racking lanes vertically connected to it. Thus, it should be needed that a shuttle must be able to move in two mutually perpendicular directions, longitudinally and transversely. In the prior art, the three-dimensional warehouse usually uses a sub-mother vehicle system, that is, a main vehicle carries a sub-vehicle and moves in the incoming and unloading lane, and then the sub-vehicle drives out in a direction perpendicular to the direction of the main vehicle. The sub-vehicle enters in racking lanes, and then goes back along the original line to the main vehicle, and finally is taken away by the main vehicle.

The existing shuttle in the type of the sub-mother vehicle system mostly uses a two-way eight-wheel structure, which has the disadvantages of low carrying capacity, weak obstacle crossing capacity and high failure rate, thus it can not meet the increasingly busy warehousing and logistics needs. Moreover, it generally uses a cam jacking structure, which has the defects of easy wear and tear, uneven force on the follower and low transmission efficiency. In addition, the existing shuttle of the sub-mother vehicle system not only occupies a considerable space, reducing the number of shelves and compartments and the space utilization of the three-dimensional warehouse, but also increases the cost of manufacturing, procurement and maintenance.

SUMMARY

Embodiments of the present disclosure provide a cable-stayed jacking reversing type sixteen-wheel four-way shuttle.

Detailed technical solutions are as follows:

The embodiments of the present disclosure provide a cable-stayed jacking reversing type sixteen-wheel four-way shuttle, comprising a support frame, a transverse movement drive mechanism, a longitudinal movement drive mechanism and a cable-stayed reversing jacking mechanism, in which the mechanisms are all mounted on the support frame, wherein:

the transverse movement drive mechanism includes transverse driving wheels, first transverse driven wheels, second transverse driven wheels and third transverse driven wheels, in which each set of the wheels is symmetrically arranged on the front and rear sides of the support frame; and, the two transverse driving wheels are respectively arranged on both ends of a transverse transmission shaft, the two third transverse

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driven wheels are respectively arranged on both ends of a transverse connecting shaft, and the transverse transmission shaft is drivingly connected to a transverse motor through a first transverse transmission belt;

the longitudinal movement drive mechanism includes first longitudinal driving wheels, second longitudinal driving wheels, first longitudinal driven wheels and second longitudinal driven wheels, in which each set of the wheels is symmetrically arranged on the left and right sides of the support frame; and, the first longitudinal driving wheels and the second longitudinal driving wheels on both sides are connected to a longitudinal transmission shaft through first longitudinal transmission belts on the corresponding side, and the longitudinal transmission shaft is drivingly connected to a longitudinal motor through a second longitudinal transmission belt; and,

the cable-stayed reversing jacking mechanism includes jacking transverse plates symmetrically arranged on the front and rear sides of the support frame, and cable-stayed jacking blocks respectively arranged at the bottoms of both ends of each jacking transverse plate; each end of the jacking transverse plate is provided with a first reversing assembling groove and a second reversing assembling groove, to assemble the transverse transmission shaft and the transverse connecting shaft; each cable-stayed jacking block is provided with a jacking shaft hole penetrating left and right and a jacking inclined hole arranging obliquely and penetrating the front and rear; a ball bearing sleeve is movably arranged in the jacking shaft hole, a cable-stayed jacking shaft is arranged between the front and rear sides of the ball bearing sleeve, and the cable-stayed jacking shaft is movably embedded in the jacking inclined hole; and, a ball screw is movably penetrated in each ball bearing sleeve, the cable-stayed jacking blocks at both ends are connected with each other through the ball screws, and the ball screws are drivingly connected to the transverse motor through a chain.

In some embodiments, the front and rear side walls of the support frame are respectively provided with four square mounting holes, and the left and right side walls of the support frame are respectively provided with four circular mounting holes; wherein:

the transverse driving wheels are connected with the transverse transmission shaft which passes through the second reversing assembling groove to assemble into the corresponding square assembling holes; the axles of the first transverse driven wheels and the second transverse driven wheels respectively pass through the corresponding square mounting holes to connect with the corresponding jacking transverse plate; and, the third transverse driven wheels are connected with the transverse connecting shaft which passes through the first reversing assembling groove to assemble into the corresponding square assembling holes; and,

the axles of the first longitudinal driving wheels and the second longitudinal driving wheels, which respectively pass through the corresponding circular mounting holes, are connected to the bottom of the supporting frame through their corresponding first bearing seats; and the first longitudinal driven wheels and the second longitudinal driven wheels are respectively assembled into the corresponding circular assembling holes.

In some embodiments, the transverse driving wheel and the first transverse driven wheel on the same side are connected with each other through a second transverse

transmission belt; and/or, the transverse driving wheel and the third transverse driven wheel on the same side are connected with each other through the second transverse transmission belt.

In some embodiments, the cable-stayed jacking reversing type sixteen-wheel four-way shuttle further comprises a tension adjusting mechanism positioned outside the first longitudinal transmission belt, wherein:

the tension adjusting mechanism includes a first tensioning pulley and a second tensioning pulley which are both attached to the outside of the first longitudinal transmission belt, the first tensioning pulley is fixedly arranged on the inner side wall of the support frame through a first fixing plate, and the second tensioning pulley is movably arranged on the inner side wall of the support frame through an adjustment plate.

In some embodiments, the adjusting plate is provided with an adjusting square hole in horizontal, and the adjusting plate is movably fixed to the inner side wall of the supporting frame by bolts passing through the adjusting square hole; and, one end of the adjusting plate is adjustably connected to a second fixing plate fixed on the inner side wall of the support frame by adjustable nuts and bolts.

In some embodiments, the cable-stayed reversing jacking mechanism further comprises:

second bearing seats, each arranged on the support frame with a bearing arranged at its top to support the corresponding ball screw; and,

jacking nuts, each fixedly sleeved on the corresponding ball bearing sleeve with its front and rear side walls respectively welded and connected to the cable-stayed jacking shaft.

In some embodiments, the cable-stayed reversing jacking mechanism further comprises:

limiting sliding grooves, each vertically arranged on the front and rear side walls of the cable-stayed jacking block; and,

limiting guide posts, each vertically arranged on the support frame and slidably connected with the corresponding limiting sliding grooves.

In some embodiments, each jacking transverse plate is slidably connected to the corresponding jacking guide posts through guide sliding grooves opened at both ends of the jacking transverse plate, and the jacking guide posts are located at the four corners of the support frame.

In some embodiments, the inclination angle of the jacking inclined hole is $5-85^\circ$, and the vertical height of the jacking inclined hole is equal to the vertical height of the jacking shaft hole.

In some embodiments, the transverse motor is respectively drivingly connected to the first transverse transmission belt and the chain through a cross reducer.

The present disclosure achieves the following beneficial effects:

- (1) In the present disclosure, the use of eight transverse wheels, eight longitudinal wheels and a reinforced support frame can improve the loading capacity; and the installation position of each wheel is optimized so as to achieve a better obstacle crossing ability;
- (2) In order to match up with sixteen wheels, the wheel transmission mechanism is optimized; and, by moving the installation position of the longitudinal transmission shaft upwards and increasing the diameter of the longitudinal wheels, the shuttle can have a better loading capacity;
- (3) The tension adjusting mechanism is used to adjust the tension of the first longitudinal transmission belt, while

ensuring the stability and continuity of the power transmission of the first longitudinal driving wheels and the second longitudinal driving wheels;

- (4) In the cable-stayed reversing jacking mechanism, the cable-stayed jacking blocks contact to the rolling line of the cable-stayed jacking shaft, with high transmission efficiency and no change in the pressure angle; and, the jacking inclined holes on both sides of the jacking block are symmetrically arranged with uniform force. The afore design is novel with compact structure, stable operation, small size and high loading capacity, thereby reducing the maintenance and maintenance times of the mechanical mechanism so as to have high work efficiency and long service life;

- (5) The transmission structure with the fitted ball screws and ball bearing sleeves makes the transmission efficiency reach 50%-70%, and the ball bearing sleeves and the jacking shaft are directly connected by jacking nuts, which reduces the loss of kinetic energy;

- (6) The sixteen-wheel four-way shuttle has a compact structure, which not only reduces costs and saves space, but also reduces the overall weight and volume of the vehicle and improves operating capacity and cargo storage efficiency.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings described here are provided for further understanding of the present disclosure, and constitute a part of the present disclosure. The exemplary embodiments of the present disclosure and illustrations thereof are intended to explain the present disclosure, but do not constitute inappropriate limitations to the present disclosure. In the drawings:

FIG. 1 is a top view diagram of a cable-stayed jacking reversing type sixteen-wheel four-way shuttle according to an embodiment of the present disclosure;

FIG. 2 is a main view diagram of a cable-stayed jacking reversing type sixteen-wheel four-way shuttle according to an embodiment of the present disclosure;

FIG. 3 is a side view diagram of a cable-stayed jacking reversing type sixteen-wheel four-way shuttle according to an embodiment of the present disclosure;

FIG. 4 is a schematic diagram of a three-dimensional structure of a cable-stayed jacking reversing type sixteen-wheel four-way shuttle according to an embodiment of the present disclosure;

FIG. 5 is a schematic diagram of a support frame according to an embodiment of the present disclosure;

FIG. 6 is an assembly structure diagram of a tension adjusting mechanism according to an embodiment of the present disclosure;

FIG. 7 is a schematic diagram of a cable-stayed reversing jacking mechanism according to an embodiment of the present disclosure;

FIG. 8 is a schematic diagram of a cable-stayed jacking block according to an embodiment of the present disclosure;

FIG. 9 is a cross-sectional schematic diagram of the cable-stayed jacking block in FIG. 8 according to an embodiment of the present disclosure;

FIG. 10 is a cross-sectional schematic diagram of a ball screw and a ball bearing sleeve according to an embodiment of the present disclosure;

FIG. 11 is a schematic diagram of a cable-stayed jacking block according to another embodiment of the present disclosure;

REFERENCE NUMERALS

100—support frame; **101**—square mounting hole(s); **102**—circular mounting hole(s); **103**—jacking guide post(s); **104**—guide post fixing plate(s); **105**—first bearing seat(s);

200—transverse movement drive mechanism; **201**—transverse driving wheel(s); **202**—first transverse driven wheel(s); **203**—second transverse driven wheel(s); **204**—third transverse driven wheel(s); **205**—second transverse transmission belt(s); **206**—transverse transmission shaft; **207**—first transverse transmission belt; **208**—transverse motor; **209**—transverse connecting shaft; **210**—cross reducer;

300—longitudinal movement drive mechanism; **301**—first longitudinal driving wheel(s); **302**—second longitudinal driving wheel(s); **303**—first longitudinal driven wheel(s); **304**—second longitudinal driven wheel(s); **305**—longitudinal transmission shaft; **306**—second longitudinal transmission belt; **307**—longitudinal motor; **308**—first longitudinal transmission belt(s);

400—cable-stayed reversing jacking mechanism; **401**—jacking transverse plate(s); **402**—cable-stayed jacking block(s); **403**—ball screw(s); **404**—second bearing seat(s); **405**—gear bearing sleeve(s); **406**—chain; **407**—ball bearing sleeve(s); **408**—jacking nut(s); **409**—cable-stayed jacking shaft(s); **410**—jacking shaft hole(s); **411**—jacking inclined hole(s); **412**—limiting sliding groove(s); **413**—limiting guide post(s); **414**—first reversing assembling groove(s); **415**—second reversing assembling groove(s); **416**—guide sliding grooves(s);

500—battery pack; **600**—controller; **700**—tension adjusting mechanism; **701**—first tensioning pulley(s); **702**—first fixing plate(s); **703**—second tensioning pulley(s); **704**—adjustment plate(s); **705**—adjusting square hole(s); **706**—second fixing plate(s).

DETAILED DESCRIPTION

To make the objectives, technical solutions, and advantages of the present disclosure clearer, the present disclosure is described below with reference to the accompanying drawings and embodiments. It should be understood that the embodiments described herein are merely used to explain the present disclosure, rather than to limit the present disclosure. All other embodiments obtained by those of ordinary skill in the art based on the embodiments of the present disclosure without creative efforts should fall within the protection scope of the present disclosure.

Apparently, the accompanying drawings in the following description show merely some embodiments of the present disclosure, and a person of ordinary skill in the art may apply the present disclosure to other similar scenarios according to these drawings without creative efforts. In addition, it can also be appreciated that, although it may take enduring and complex efforts to achieve such a development process, for those of ordinary skill in the art related to the present disclosure, some changes such as design, manufacturing or production made based on the technical content in the present disclosure are merely regular technical means, and should not be construed as insufficiency of the present disclosure.

The “embodiment” mentioned in the present disclosure means that a specific feature, structure, or characteristic described in combination with the embodiment may be included in at least one embodiment of the present disclosure.

The phrase appearing in different parts of the specification does not necessarily refer to the same embodiment or an independent or alternative embodiment exclusive of other embodiments. It may be explicitly or implicitly appreciated by those of ordinary skill in the art that the embodiment described herein may be combined with other embodiments as long as no conflict occurs.

Unless otherwise defined, the technical or scientific terms used in the present disclosure are as they are usually understood by those of ordinary skill in the art to which the present disclosure pertains. The terms “one”, “a”, “the” and similar words are not meant to be limiting, and may represent a singular form or a plural form. The terms “include”, “contain”, “have” and any other variants in the present disclosure mean to cover the non-exclusive inclusion, for example, a process, method, system, product, or device that includes a series of steps or modules (units) is not necessarily limited to those steps or units which are clearly listed, but may include other steps or units which are not expressly listed or inherent to such a process, method, system, product, or device. “Connected”, “interconnected”, “coupled” and similar words in the present disclosure are not restricted to physical or mechanical connections, but may include electrical connections, whether direct or indirect. The term “multiple” in the present disclosure means two or more. The term “and/or” describes associations between associated objects, and it indicates three types of relationships. For example, “A and/or B” may indicate that A exists alone, A and B coexist, or B exists alone. The character “/” generally indicates that the associated objects are in an “or” relationship. The terms “first”, “second”, “third” and so on in the present disclosure are intended to distinguish between similar objects but do not necessarily indicate a specific order of the objects.

Referring to FIG. 1, an embodiment of the present disclosure provides a cable-stayed jacking reversing type sixteen-wheel four-way shuttle, comprising a support frame **100**, a transverse movement drive mechanism **200**, a longitudinal movement drive mechanism **300**, a cable-stayed reversing jacking mechanism **400**, a battery pack **500** and a controller **600**, in which the mechanisms, the battery pack and the controller are all mounted on the support frame **100**.

In one example, the supporting frame **100** is an integral frame formed by the connecting between a bending sheet metal and local reinforcing ribs, which greatly improves the carrying capacity of the vehicle. The battery pack **500** is used as a power source of the vehicle, and the battery pack **500** can use lithium-ion rechargeable batteries, storage batteries or any other energy storage batteries to supply power for each function unit of the vehicle. The controller **600** is a key core component of the whole vehicle, which controls the movement of the vehicle. The controller **600** is electrically connected to the battery pack **500**, a transverse motor **208** of the transverse movement drive mechanism **200**, a longitudinal motor **307** of the longitudinal movement mechanism **300** and a jacking motor **408** of the cable-stayed reversing jacking mechanism **400**. The flexibly longitudinal or transverse movement of the vehicle is controlled by the controller **600**, and the jacking and reversing of the vehicle is achieved by the cable-stayed reversing jacking mechanism **400**.

In some preferred embodiments, referring to FIG. 2 and FIG. 4, the transverse movement drive mechanism **200** acts as an execution unit for a transverse movement of the sixteen-wheel four-way shuttle. The transverse movement drive mechanism **200** includes transverse driving wheels **201**, first transverse driven wheels **202**, second transverse driven wheels **203** and third transverse driven wheels **204**, in

which each set of the wheels is symmetrically arranged on the front and rear sides of the support frame 100. Each set of the wheels has two wheels symmetrically arranged on the front and rear sides of the support frame 100 to form a structure of a one-way eight-wheel vehicle, so as to improve the carrying capacity and the obstacle crossing ability of the vehicle during moving transversely.

In one example, referring to FIG. 2 and FIG. 4, the two transverse driving wheels 201 are respectively arranged on both ends of a transverse transmission shaft 206. The transverse driving wheels 201 at both ends can be driven to rotate through the transverse transmission shaft 206, and the transverse transmission shaft 206 is provided with a transverse coupling. In order to improve the moving stability of the vehicle, the two third transverse driven wheels 204 are connected through a transverse connecting shaft 209 to ensure that the third transverse driven wheels 204 on both sides can run synchronously. Additionally, the transverse transmission shaft 206 is connected to the transverse motor 208 through a first transverse transmission belt 207, the transverse transmission shaft 206 is driven to rotate by the transverse motor 208 through the first transverse transmission belt 207, and the transverse motor 208 is a servo drive motor.

The working principle of the transverse movement drive mechanism 200 in the sixteen-wheel four-way shuttle is as follows: the transverse transmission shaft 206 is driven to rotate by the transverse motor 208 through the first transverse transmission belt 207, the transverse driving wheels 201 at both ends of the transverse transmission shaft 206 are simultaneously driven to rotate; and, the first transverse driven wheels 202, the second transverse driven wheels 203 and the third transverse driven wheels 204 on the support frame 100 are synchronously driven to rotate, so as to realize the transverse moving of the sixteen-wheel four-way shuttle.

In some preferred embodiments, referring to FIG. 3 and FIG. 4, the longitudinal movement drive mechanism 300 acts as an execution unit for a longitudinal movement of the sixteen-wheel four-way shuttle. The longitudinal movement drive mechanism 300 includes first longitudinal driving wheels 301, second longitudinal driving wheels 302, first longitudinal driven wheels 303 and second longitudinal driven wheels 304, in which each set of the wheels is symmetrically arranged on the left and right sides of the support frame 100. Each set of the wheels has two wheels symmetrically arranged on both sides of the support frame 100 to form a structure of a one-way eight-wheel vehicle, so as to improve the carrying capacity and the obstacle crossing ability of the vehicle during moving longitudinally.

In one example, referring to FIG. 3 and FIG. 4, the first longitudinal driving wheels 301 and the second longitudinal driving wheels 302 on both sides are used as driving wheels and are respectively connected to a longitudinal transmission shaft 305 through first longitudinal transmission belts 308 on both sides, and the longitudinal transmission shaft 305 is connected to the longitudinal motor 307 through a second longitudinal transmission belt 306. The longitudinal transmission shaft 305 is driven to rotate by the longitudinal motor 307 through the second longitudinal transmission belt 306, and the longitudinal motor 307 is a servo drive motor.

In one example, in order to match sixteen wheels, the transmission mechanism of the longitudinal wheels is optimized. The position of the longitudinal transmission shaft 305 is moved up, so that the assembling structure of the longitudinal transmission shaft 305, the first longitudinal driving wheels 301 and the second longitudinal driving wheels 302 can form a triangular structure on its side. On the

one hand, the first longitudinal driving wheels 301 and the second longitudinal driving wheels 302 are driven to rotate synchronously through the longitudinal transmission shaft 305. On the other hand, the longitudinal transmission shaft 305 arranged in an elevated position can be well staggered from the reversing jacking mechanism 400 arranged below it, which makes the structure of the vehicle more compact, reasonable and space-saving.

In one example, in order to make the vehicle have a better carrying capacity, the diameter of the eight longitudinal wheels used in the longitudinal movement drive mechanism 300 is larger than the diameter of the eight transverse wheels used in the transverse movement drive mechanism 200. Specifically, the diameters of the transverse driving wheels 201, the first transverse driven wheels 202, the second transverse driven wheels 203 and the third transverse driven wheels 204 are all slightly smaller than the diameters of the first longitudinal driving wheels 301, the second longitudinal driving wheels 302, the first longitudinal driven wheels 303 and the second longitudinal driven wheels 304. The diameter ratio is about 1:1.1-1:1.3; preferably, 1:1.2.

The working principle of the longitudinal movement drive mechanism 300 in the sixteen-wheel four-way shuttle is as follows: the longitudinal transmission shaft 305 is driven to rotate by the longitudinal motor 307 through the second longitudinal transmission belt 306, the first longitudinal driving wheels 301 and the second longitudinal driving wheels 302 at both ends of the longitudinal transmission shaft 305 are simultaneously driven to rotate; and, the first longitudinal driven wheels 303 and the second longitudinal driven wheels 304 on the support frame 100 are synchronously driven to rotate, so as to realize the longitudinal moving of the sixteen-wheel four-way shuttle. Additionally, a longitudinal coupling is arranged on the longitudinal transmission shaft 305. Moreover, the second longitudinal transmission belt 306 will be re-tensioned after working for a period of time, and the adjustment can be achieved by a transverse movement of the tension adjustment mechanism 700 according to the tightness of the pulley.

In some preferred embodiments, referring to FIG. 4, FIG. 7, FIG. 8, FIG. 9, FIG. 10 and FIG. 11, the cable-stayed reversing jacking mechanism 400 acts as an execution unit for a jacking and reversing movement of the sixteen-wheel four-way shuttle. The cable-stayed reversing jacking mechanism 400 includes jacking transverse plates 401 symmetrically arranged on the front and rear sides of the support frame 100, and cable-stayed jacking blocks 402 respectively arranged at the bottoms of both ends of each jacking transverse plate 401; each end of the jacking transverse plate 401 is provided with a first reversing assembling groove 414 and a second reversing assembling groove 415, to assemble the transverse transmission shaft 206 and the transverse connecting shaft 209; each cable-stayed jacking block 402 is provided with a jacking shaft hole 410 penetrating left and right and a jacking inclined hole 411 arranging obliquely and penetrating the front and rear; a ball bearing sleeve 407 is movably arranged in the jacking shaft hole 410; a cable-stayed jacking shaft 409 is arranged between the front and rear sides of the ball bearing sleeve 407, and the cable-stayed jacking shaft 409 is movably embedded in the jacking inclined hole 411; and, a ball screw 403 is movably penetrated in each ball bearing sleeve 407, the cable-stayed jacking blocks 402 at both ends are connected with each other through the ball screws 403, and the ball screws 403 are drivingly connected to the transverse motor 208 through a chain 406.

The working principle of the cable-stayed reversing jacking mechanism **400** in the sixteen-wheel four-way shuttle is as follows: the ball screws **403** on the front and rear sides are driven to rotate synchronously by the transverse motor **208** through the chain **406**. During the rotation of the ball screws **403**, the ball bearing sleeve **407** and the two cable-stayed jacking shafts **409** are driven to slide left and right along the jacking shaft hole **410** and the jacking inclined hole **411** respectively, and the cable-stayed jacking block **402** is driven to move up and down. Then the jacking transverse plates **401** on the top of the cable-stayed jacking shaft **409** and the jacking platform provided on the jacking transverse plates **401** are driven to lift up and down, thus realizing the functions of jacking and reversing.

In one example, referring to FIG. 4 and FIG. 7, two ends of the transverse transmission shaft **206** and two ends of the transverse connection shaft **209** are respectively arranged passing through the first reversing assembling grooves **414** and the second reversing assembling grooves **415** at both ends of the corresponding jacking transverse plate **401**, so that the transverse transmission shaft **206** and the transverse connection shaft **209** can move up and down synchronously along with the jacking transverse plate **401**. Then, the eight transverse wheels of the transverse drive mechanism **200**, consisting of the transverse driving wheels **201**, the first transverse driven wheels **202**, the second transverse driven wheels **203** and the third transverse driven wheels **204**, are driven to move up and down synchronously relative to the eight longitudinal wheels, so as to leave the ground or contact the ground, thereby realizing the reversing function between the transverse movement drive mechanism **200** and the longitudinal movement drive mechanism **300**.

In one example, referring to FIG. 7 and FIG. 9, the cable-stayed jacking blocks **402** on the same side are linked by the ball screws **403**, and the two cable-stayed jacking blocks **402** are respectively arranged on both ends of the ball screws **403**. The middle position of each ball screw **403** is sleeved with a gear bearing sleeve **405**, and the gear bearing sleeve **405** is connected to the transverse motor **208** through the chain **406** and the cross reducer **210**. After the transverse motor **208** is decelerated and reversed by the cross reducer **210**, the torque of the ball screw **403** is controlled after being driven by the chain **406**. Then the cable-stayed jacking blocks **402** at both ends are driven to lift up and down by the ball screw **403**, so that the jacking transverse plates **401** and the jacking platform provided on the jacking transverse plates **401** are driven to lift up and down, thus realizing the functions of jacking and reversing.

In some preferred embodiments, referring to FIG. 5, the supporting frame **100** is an integral frame formed by the connecting between a bending sheet metal and local reinforcing ribs. The front and rear side walls of the support frame **100** are respectively provided with four square mounting holes **101** to assemble the transverse wheels, and correspondingly, the left and right side walls of the support frame **100** are respectively provided with four circular mounting holes **102** to assemble the longitudinal wheels.

In one example, referring to FIG. 5 and FIG. 7, the transverse driving wheels **201** on the front and rear sides are connected with the transverse transmission shaft **206** which passes through the second reversing assembling groove **415** to assemble into the corresponding square assembling holes **101**; the axles of the first transverse driven wheels **202** and the second transverse driven wheels **203** respectively pass through the corresponding square mounting holes **101** to connect with the corresponding jacking transverse plates **401**; and, the third transverse driven wheels **204** are con-

nected with the transverse connecting shaft **209** which passes through the first reversing assembling groove **414** to assemble into the corresponding square assembling holes **101**. Moreover, in order to cooperate with the jacking and reversing functions of the reversing jacking mechanism **400**, the transverse driving wheels **201**, the first transverse driven wheels **202**, the second transverse driven wheels **203** and the third transverse driven wheels **204** along with the jacking transverse plates **401** can be lifted between the upper and lower limit positions in the corresponding square assembling holes **101**.

In one example, referring to FIG. 4, FIG. 5 and FIG. 6, the axles of the first longitudinal driving wheels **301** and the second longitudinal driving wheels **302**, which respectively pass through the corresponding circular mounting holes **102**, are connected to the longitudinal transmission shaft **305**. The first longitudinal driving wheels **301** and the second longitudinal driving wheels **302** are driven to rotate synchronously by the longitudinal transmission shaft **305** through the first longitudinal belts **308** to improve the carrying capacity of the vehicle. In addition, the axles of the first longitudinal driving wheels **301** and the second longitudinal driving wheels **302** are respectively arranged passing through the corresponding circular assembling holes **102** and are fixedly connected to the bottom of the support frame **100** through first bearing seats **105**, so as to ensure the stability of the first longitudinal driving wheels **301** and the second longitudinal driving wheels **302** during rotating.

In one example, referring to FIG. 3, FIG. 4, FIG. 5 and FIG. 6, the first longitudinal driven wheels **303** and the second longitudinal driven wheels **304** are used as driven wheels, and are driven passively by the support frame **100** to mainly have the function of supporting and guiding. The first longitudinal driven wheels **303** and the second longitudinal driven wheels **304** are fixed in the corresponding circular mounting holes **102**.

In some preferred embodiments, referring to FIG. 2, FIG. 4 and FIG. 7, the wheel installation structure of the sixteen-wheel four-way shuttle is optimized. The first transverse driven wheels **202** are used as the auxiliary driving wheels of the vehicle, which further improves the transverse gripping ability and obstacle crossing ability of the vehicle. Specifically, the transverse driving wheel **201** and the first transverse driven wheel **202** on the same side are connected with each other through the second transverse transmission belt **205**. The power of the transverse driving wheels **201** is transmitted to the first transverse driven wheels **202** through the second transverse transmission belt **205**, and the first transverse driven wheels **202** can be driven to rotate synchronously.

In addition, according to the actual operation requirements of the vehicle, the transverse driving wheel **201** and the third transverse driven wheel **204** on the same side are connected with each other through the second transverse transmission belt **205**. The third transverse driven wheels **204** are used as a linkage mechanism of the transverse driving wheels **201**, thereby further improving the stability, the ground gripping ability and obstacle crossing ability during the transverse moving of the vehicle.

In some preferred embodiments, referring to FIG. 1 and FIG. 5, in order to ensure the stability of the power transmission of the longitudinal movement drive mechanism **300**, the sixteen-wheel four-way shuttle further comprises a tension adjusting mechanism **700** positioned outside each of the first longitudinal transmission belt **308**. The connection between the first longitudinal transmission belts **308**, the first longitudinal driving wheels **301** and the second longi-

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tudinal driving wheels **302** is optimized through the tension adjustment mechanism **700**, so that the installation of the first longitudinal transmission belts **308** can be more convenient.

In one example, referring to FIG. 6, each tension adjusting mechanism **700** includes a first tensioning pulley **701** and a second tensioning pulley **703** which are both attached to the outside of the corresponding first longitudinal transmission belt **308**. The first tensioning pulley **701** is fixedly arranged on the inner side wall of the support frame **100** through a first fixing plate **702**. The first tensioning pulley **701** is fixed to be always located between the first longitudinal driving wheels **301** and the longitudinal transmission shaft **305**, and to be always in contact with the first longitudinal transmission belt **308**. The second tensioning pulley **703** is movably arranged on the inner side wall of the support frame **100** through an adjustment plate **704**. With the adjusting function of the adjusting plate **704** to the second tensioning pulley **703**, the second tensioning pulley **703** can have a relative displacement to the longitudinal transmission shaft **305**, so as to press tightly to fit against the other side end surface of the first longitudinal transmission belt **308** to form a tensioned state for the first longitudinal transmission belt **308**. Alternatively, the second tensioning pulley **703** moves away from the longitudinal transmission shaft **305** to separate it from the first longitudinal transmission belt **308**, thus the tensioned state of the first longitudinal transmission belt **308** can be eliminated.

In one example, referring to FIG. 6, the adjusting plate **704** is provided with an adjusting square hole **705** in horizontal. The adjusting plate **704**, passing through the adjusting square hole **705**, is movably fixed to the inner side wall of the supporting frame **100** by bolts. The adjusting square hole **705** has a certain limiting function under the action of the bolts. The left and right adjustable spacing of the second tensioning pulley **703** is determined by the length of the adjusting square hole **705**, to meet the tensioning requirement of the first longitudinal transmission belt **308**.

In one example, referring to FIG. 6, a distal end of the adjusting plate **704** is adjustably connected to a second fixing plate **706** by adjustable nuts and bolts. In the presence of adjusting nuts and bolts, the adjusting plate **704** and the second tensioning pulley **703** arranged on the adjusting plate **704** can be controlled to move left and right along the length direction of the adjusting square hole **705**, thereby adjusting the tension of the first longitudinal transmission belt **308**. The second fixing plate **706** is fixed on the inner side wall of the supporting frame **100**. The adjusting plate **704** and the second fixing plate **706** are both L-shaped plates, and their side end surfaces are arranged oppositely. The adjusting nuts and bolts are located on the oppositely arranged L-shaped plates. The adjustment of the adjusting nuts and bolts can be achieved by the existing conventional structure, which will not be detailed described here.

In one example, referring to FIG. 4, FIG. 7, FIG. 8 and FIG. 9, the cable-stayed reversing jacking mechanism **400** further comprises second bearing seats **404**, each of which is arranged on the support frame **100** with a bearing arranged at its top to support the corresponding ball screw **403**. There are at least two second bearing seats **404** on each side of the support frame **100**, which are respectively disposed between and/or on both sides of the cable-stayed jacking blocks **402**. Each ball screw **403** is installed in a mounting hole on the upper end of the corresponding second bearing seat **404** through a bearing. The two ends of the ball screw **403** both protrude out of the cable-stayed jacking block **402** and pass through the bearing installed in the mounting hole. The

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second bearing seat **404** has the functions of stabilizing and supporting the ball screw **403**, thus improving the stability of the operation of the cable-stayed reversing jacking assembly.

In one example, referring to FIG. 4, FIG. 7, FIG. 9 and FIG. 11, the cable-stayed reversing jacking mechanism **400** further comprises jacking nuts **408**, each of which is fixedly sleeved on the ball bearing sleeve **407** with its front and rear side walls respectively welded and connected to the cable-stayed jacking shaft **409**. The jacking nut **408** is a copper nut, and each jacking nut **408** is fixedly sleeved on the corresponding ball bearing sleeve **407** with a 50%-70% transmission efficiency, which reduces the kinetic energy loss.

In one example, referring to FIG. 8 and FIG. 9, the cable-stayed reversing jacking mechanism **400** further comprises limiting sliding grooves **412** and limiting guide posts **413**, in which the limiting sliding grooves **412** are respectively vertically arranged on the front and rear side walls of the cable-stayed jacking block **402**, and the limiting guide posts **413** are vertically arranged on the support frame **100** and slidably connected with the limiting sliding grooves **412**. The limiting guide posts **413** are used to limit the horizontal movement of the cable-stayed jacking blocks **402** and play a role of vertical guidance simultaneously. During the movement of the ball bearing sleeve **307**, in order to prevent the cable-stayed jacking blocks **402** from moving due to the horizontal pulling force, the horizontal pulling force received by the cable-stayed jacking blocks **402** is converted to the vertical pulling force through the jacking inclined holes **411**, so that the cable-stayed jacking blocks **402** can move up and down along the limiting guide posts **413**.

In one example, referring to FIG. 4 and FIG. 7, in order to ensure the stability of the cable-stayed reversing jacking mechanism **400** during the jacking and reversing process, each jacking transverse plate **401** is connected to corresponding jacking guide posts **103** by sliding up and down through guide sliding grooves **416**, the guide sliding grooves **416** are opened at both ends of the jacking transverse plate **401**, and the jacking guide posts **103** are located at the four corners of the support frame **100**. Both ends of the jacking transverse plate **401** can slide along the jacking guide posts **103** during lifting, to avoid the defect that the jacking transverse plate **401** sways from side to side during jacking. The upper and lower ends of the jacking guide post **103** are fixedly connected to the support frame **100** through guide post fixing plates **104** respectively. The guide post fixing plates **104** provided at the upper and lower ends simultaneously play a limiting role in the lifting and lowering of the jacking transverse plate **401**.

In one example, referring to FIG. 11, according to the implementation requirements, different inclination angles of the jacking inclined hole **411** can be designed for different working conditions. The inclination angle of the jacking inclined hole **411** is 5-85°; preferably, 15-75°; more preferably, 25-65°; further preferably, 30-65°; furthermore preferably, 35-65°; most preferably, 40-45°.

In one example, referring to FIG. 9 and FIG. 11, the jacking shaft hole **410** is communicated with the jacking inclined hole **411**. The shape of the jacking shaft hole **410** and the jacking inclined hole **411** is designed in the structure of an arc slot hole. The vertical height of the jacking shaft hole **410** is equal to the vertical height of the jacking inclined hole **411**, so as to synchronously satisfy the requirement of the up and down movement of the ball screw **403** in the jacking shaft hole **410** and the up and down movement of the

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cable-stayed jacking shaft **409** in the jacking inclined hole **411**, during the up and down moving of the jacking blocks.

The cable-stayed jacking reversing type sixteen-wheel four-way shuttle in the present disclosure uses eight wheels on each side to form a structure of a sixteen-wheel two-way moving vehicle, which greatly improves the carrying capacity and obstacle crossing ability of the vehicle. Using the cable-stayed reversing jacking mechanism, the cable-stayed jacking blocks contact the rolling line of the cable-stayed jacking shaft, which makes the transmission efficiency high with an un-changed pressure angle. The jacking inclined holes on both sides of the jacking block are symmetrically arranged with a uniform force. The structure of the vehicle is novel and compact with stable operation, small size and high carrying capacity, which reduces the maintenance and maintenance times of the mechanical mechanism so as to have a high working efficiency and a long service life. The cable-stayed jacking reversing type sixteen-wheel four-way shuttle has a compact structure, which not only reduces costs and saves space, but also reduces the overall weight and volume of the vehicle and improves the running capacity and cargo storage efficiency.

Those skilled in the art should understand that, the technical features of the above embodiments can be arbitrarily combined. In an effort to provide a concise description, not all possible combinations of all the technical features of the embodiments are described. However, these combinations of technical features should be construed as disclosed in the description as long as no contradiction occurs.

The above embodiments are merely illustrative of several implementation manners of the present disclosure, and the description thereof is more specific and detailed, but is not to be construed as a limitation to the patentable scope of the present disclosure. It should be pointed out that several variations and improvements can be made by those of ordinary skill in the art without departing from the conception of the present disclosure, but such variations and improvements should fall within the protection scope of the present disclosure. Therefore, the protection scope of the present disclosure should be subject to the protection scope defined by the claims.

What is claimed is:

1. A cable-stayed jacking reversing type sixteen-wheel four-way shuttle, characterized in that the cable-stayed jacking reversing type sixteen-wheel four-way shuttle comprising a support frame (**100**), a transverse movement drive mechanism (**200**), a longitudinal movement drive mechanism (**300**) and a cable-stayed reversing jacking mechanism (**400**), in which the mechanisms are all mounted on the support frame (**100**), wherein:

the transverse movement drive mechanism (**200**) includes transverse driving wheels (**201**), first transverse driven wheels (**202**), second transverse driven wheels (**203**) and third transverse driven wheels (**204**), in which each set of the wheels is symmetrically arranged on the front and rear sides of the support frame (**100**); and, the two transverse driving wheels (**201**) are respectively arranged on both ends of a transverse transmission shaft (**206**), the two third transverse driven wheels (**204**) are respectively arranged on both ends of a transverse connecting shaft (**209**), and the transverse transmission shaft (**206**) is drivingly connected to a transverse motor (**208**) through a first transverse transmission belt (**207**);

the longitudinal movement drive mechanism (**300**) includes first longitudinal driving wheels (**301**), second longitudinal driving wheels (**302**), first longitudinal

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driven wheels (**303**) and second longitudinal driven wheels (**304**), in which each set of wheels is symmetrically arranged on the left and right sides of the support frame (**100**); and, the first longitudinal driving wheels (**301**) and the second longitudinal driving wheel (**302**) on both sides are connected to a longitudinal transmission shaft (**305**) through first longitudinal transmission belts (**308**) on the corresponding side, and the longitudinal transmission shaft (**305**) is drivingly connected to a longitudinal motor (**307**) through a second longitudinal transmission belt (**306**); and,

the cable-stayed reversing jacking mechanism (**400**) includes jacking transverse plates (**401**) symmetrically arranged on the front and rear sides of the support frame (**100**), and cable-stayed jacking blocks (**402**) respectively arranged at the bottoms of both ends of each jacking transverse plate (**401**); each end of the jacking transverse plate (**401**) is provided with a first reversing assembling groove (**414**) and a second reversing assembling groove (**415**), to assemble the transverse transmission shaft (**206**) and the transverse connecting shaft (**209**); each cable-stayed jacking block (**402**) is provided with a jacking shaft hole (**410**) penetrating left and right and a jacking inclined hole (**411**) arranging obliquely and penetrating the front and rear; a ball bearing sleeve (**407**) is movably arranged in the jacking shaft hole (**410**), a cable-stayed jacking shaft (**409**) is arranged between the front and rear sides of the ball bearing sleeve (**407**), and the cable-stayed jacking shaft (**409**) is movably embedded in the jacking inclined hole (**411**); and, a ball screw (**403**) is movably penetrated in each ball bearing sleeve (**407**), the cable-stayed jacking blocks (**402**) at both ends are connected with each other through the ball screws (**403**), and the ball screws (**403**) are drivingly connected to the transverse motor (**208**) through a chain (**406**).

2. The cable-stayed jacking reversing type sixteen-wheel four-way shuttle according to claim 1, wherein the front and rear side walls of the support frame (**100**) are respectively provided with four square mounting holes (**101**), and the left and right side walls of the support frame (**100**) are respectively provided with four circular mounting holes (**102**);

the transverse driving wheels (**201**) are connected with the transverse transmission shaft (**206**) which passes through the second reversing assembling groove (**415**) to assemble into the corresponding square assembling holes (**101**); the axles of the first transverse driven wheels (**202**) and the second transverse driven wheels (**203**) respectively pass through the corresponding square mounting holes (**101**) to connect with the corresponding jacking transverse plate (**401**); and, the third transverse driven wheels (**204**) are connected with the transverse connecting shaft (**209**) which passes through the first reversing assembling groove (**414**) to assemble into the corresponding square assembling holes (**101**); and,

the axles of the first longitudinal driving wheels (**301**) and the second longitudinal driving wheels (**302**), which respectively pass through the corresponding circular mounting holes (**102**), are connected to the bottom of the supporting frame (**100**) through their corresponding first bearing seats (**105**); and the first longitudinal driven wheels (**303**) and the second longitudinal driven wheels (**304**) are respectively assembled into the corresponding circular assembling holes (**102**).

3. The cable-stayed jacking reversing type sixteen-wheel four-way shuttle according to claim 1, wherein the trans-

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verse driving wheel (201) and the first transverse driven wheel (202) on the same side are connected with each other through a second transverse transmission belt (205); and/or, the transverse driving wheel (201) and the third transverse driven wheel (204) on the same side are connected with each other through the second transverse transmission belt (205).

4. The cable-stayed jacking reversing type sixteen-wheel four-way shuttle according to claim 1, further comprising a tension adjusting mechanism (700) positioned outside the first longitudinal transmission belt (308), wherein:

the tension adjusting mechanism (700) includes a first tensioning pulley (701) and a second tensioning pulley (703) which are both attached to the outside of the first longitudinal transmission belt (308), the first tensioning pulley (701) is fixedly arranged on the inner side wall of the support frame (100) through a first fixing plate (702), and the second tensioning pulley (703) is movably arranged on the inner side wall of the support frame (100) through an adjustment plate (704).

5. The cable-stayed jacking reversing type sixteen-wheel four-way shuttle according to claim 4, wherein the adjusting plate (704) is provided with an adjusting square hole (705) in horizontal, and the adjusting plate (704) is movably fixed to the inner side wall of the supporting frame (100) by bolts passing through the adjusting square hole (705); and, one end of the adjusting plate (704) is adjustably connected to a second fixing plate (706) fixed on the inner side wall of the support frame (100) by adjustable nuts and bolts.

6. The cable-stayed jacking reversing type sixteen-wheel four-way shuttle according to claim 1, wherein the cable-stayed reversing jacking mechanism (400) further comprises:

second bearing seats (404), each arranged on the support frame (100) with a bearing arranged at its top to support the corresponding ball screw (403); and,

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jacking nuts (408), each fixedly sleeved on the corresponding ball bearing sleeve (407) with its front and rear side walls respectively welded and connected to the cable-stayed jacking shaft (409).

7. The cable-stayed jacking reversing type sixteen-wheel four-way shuttle according to claim 6, wherein the cable-stayed reversing jacking mechanism (400) further comprises:

limiting sliding grooves (412), each vertically arranged on the front and rear side walls of the cable-stayed jacking block (402); and,

limiting guide posts (413), each vertically arranged on the support frame (100) and slidably connected with the corresponding limiting sliding groove (412).

8. The cable-stayed jacking reversing type sixteen-wheel four-way shuttle according to claim 1, wherein each jacking transverse plate (401) is slidably connected to corresponding jacking guide posts (103) through guide sliding grooves (416) opened at both ends of the jacking transverse plate (401), and the jacking guide posts (103) are located at the four corners of the support frame (100).

9. The cable-stayed jacking reversing type sixteen-wheel four-way shuttle according to claim 1, wherein the inclination angle of the jacking inclined hole (411) is 5-85°, and the vertical height of the jacking inclined hole (411) is equal to the vertical height of the jacking shaft hole (410).

10. The cable-stayed jacking reversing type sixteen-wheel four-way shuttle according to claim 1, wherein the transverse motor (208) is respectively drivingly connected to the first transverse transmission belt (207) and the chain (406) through a cross reducer (210).

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